

NIM : 245410034

Nama : Irvan Cahya Nugraha

Kelas : IF-1

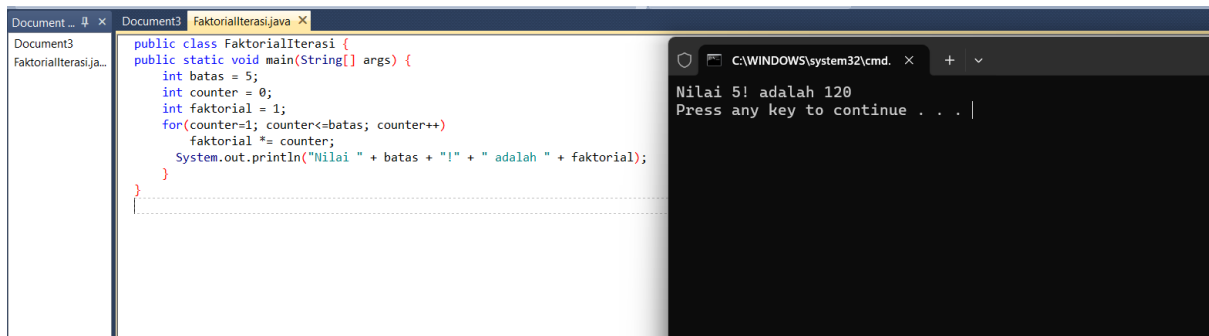
LISTING PRAKTIKUM ALGORITMA DAN PEMROGRAMAN LANJUT

MODUL 12

REKRUSIF vs ITERASI

PRAKTIK

Praktik 1

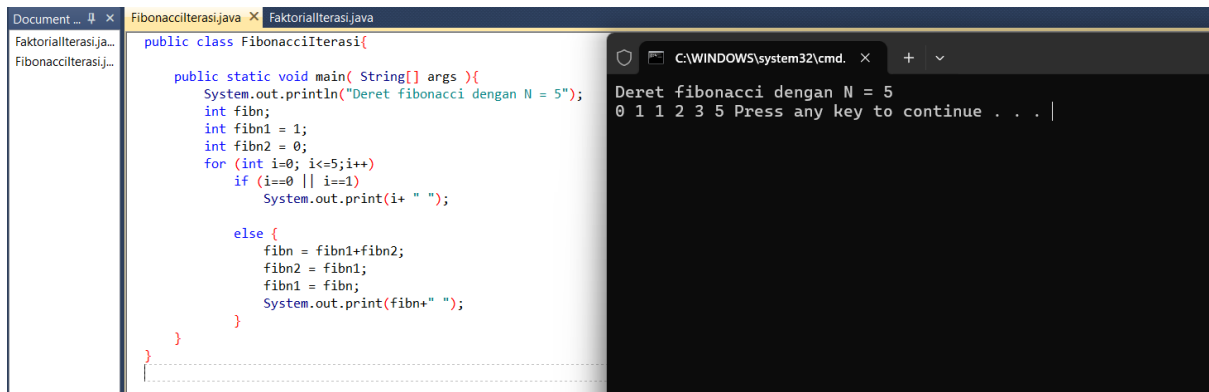


The screenshot shows an IDE with a file named `FaktorialIterasi.java`. The code defines a class `FaktorialIterasi` with a `main` method that calculates the factorial of a number using a loop. The output window shows the result for `N=5`.

```
public class FaktorialIterasi {  
    public static void main(String[] args) {  
        int batas = 5;  
        int counter = 0;  
        int faktorial = 1;  
        for(counter=1; counter<=batas; counter++)  
            faktorial *= counter;  
        System.out.println("Nilai " + batas + "!" + " adalah " + faktorial);  
    }  
}
```

Output: Nilai 5! adalah 120
Press any key to continue . . . |

Praktik 2

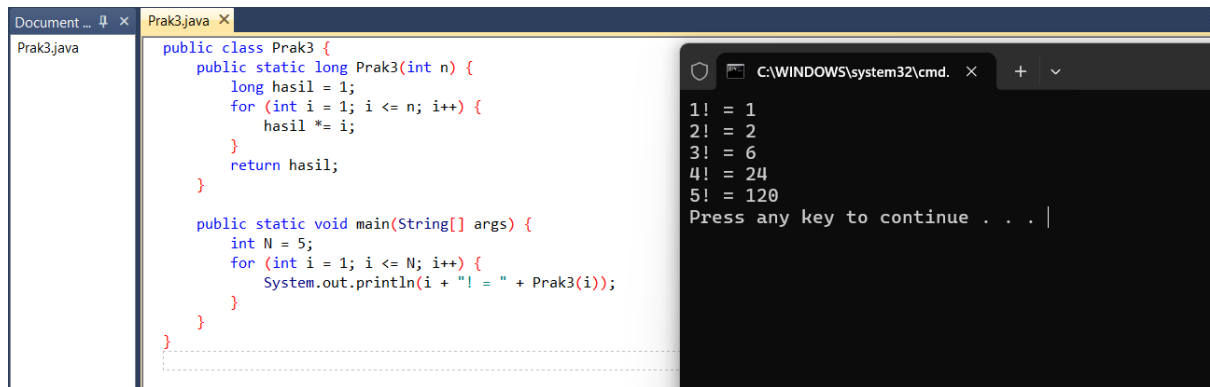


The screenshot shows an IDE with a file named `FibonacciIterasi.java`. The code defines a class `FibonacciIterasi` with a `main` method that prints the first `N` numbers of the Fibonacci sequence using a loop. The output window shows the result for `N=5`.

```
public class FibonacciIterasi {  
    public static void main( String[] args ){  
        System.out.println("Deret fibonacci dengan N = 5");  
        int fibn;  
        int fibn1 = 1;  
        int fibn2 = 0;  
        for (int i=0; i<=5;i++)  
            if (i==0 || i==1)  
                System.out.print(i+ " ");  
            else {  
                fibn = fibn1+fibn2;  
                fibn2 = fibn1;  
                fibn1 = fibn;  
                System.out.print(fibn+" ");  
            }  
    }  
}
```

Output: Deret fibonacci dengan N = 5
0 1 1 2 3 5 Press any key to continue . . . |

Praktik 3

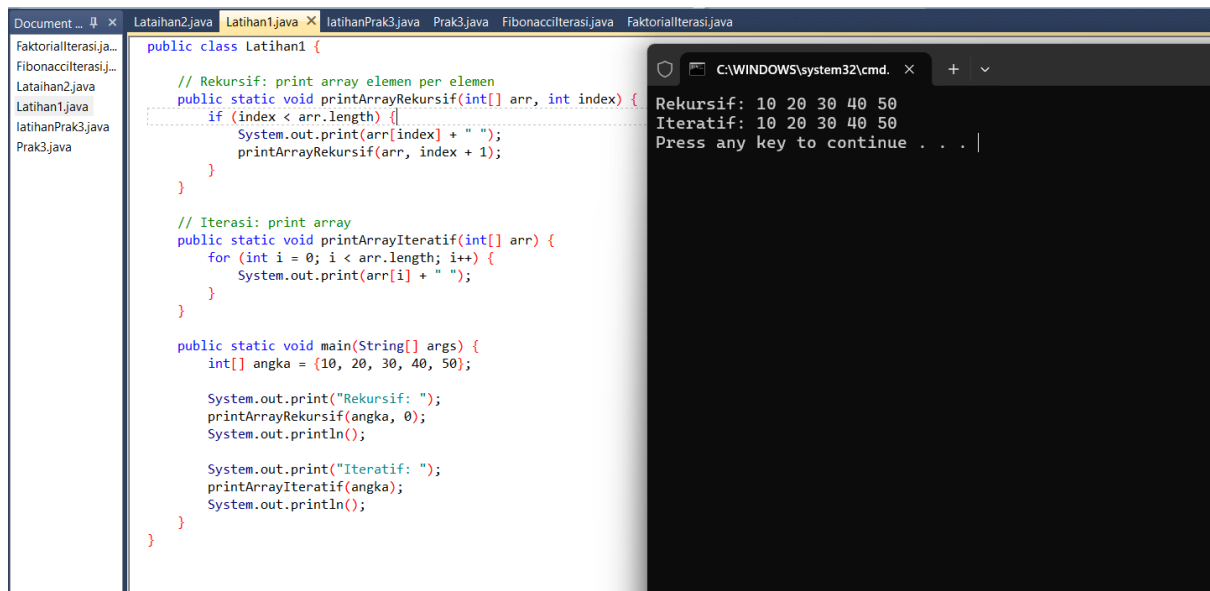


The screenshot shows an IDE with a file named `Prak3.java` open. The code defines a class `Prak3` with two methods: `Prak3(int n)` which calculates the factorial of `n` using a loop, and `main(String[] args)` which sets `N = 5` and prints the results of `Prak3(i)` for `i` from 1 to 5. To the right, a command prompt window shows the output of the program:

```
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
Press any key to continue . . . |
```

LATIHAN

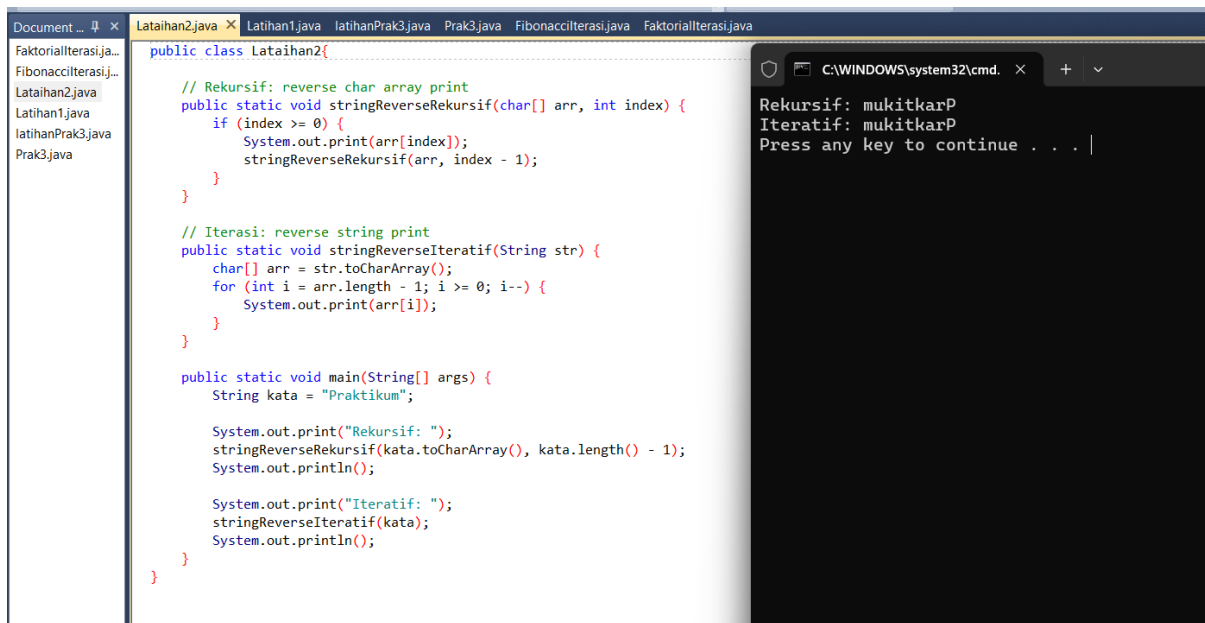
Latihan 1



The screenshot shows an IDE with a file named `Latihan1.java` open. The code defines a class `Latihan1` with two methods: `printArrayRekursif(int[] arr, int index)` which prints the elements of the array using recursion, and `printArrayIteratif(int[] arr)` which prints the elements of the array using a loop. The `main` method sets an array `angka = {10, 20, 30, 40, 50}` and calls both `printArrayRekursif` and `printArrayIteratif` on this array. To the right, a command prompt window shows the output of the program:

```
Rekursif: 10 20 30 40 50
Iteratif: 10 20 30 40 50
Press any key to continue . . . |
```

Latihan 2



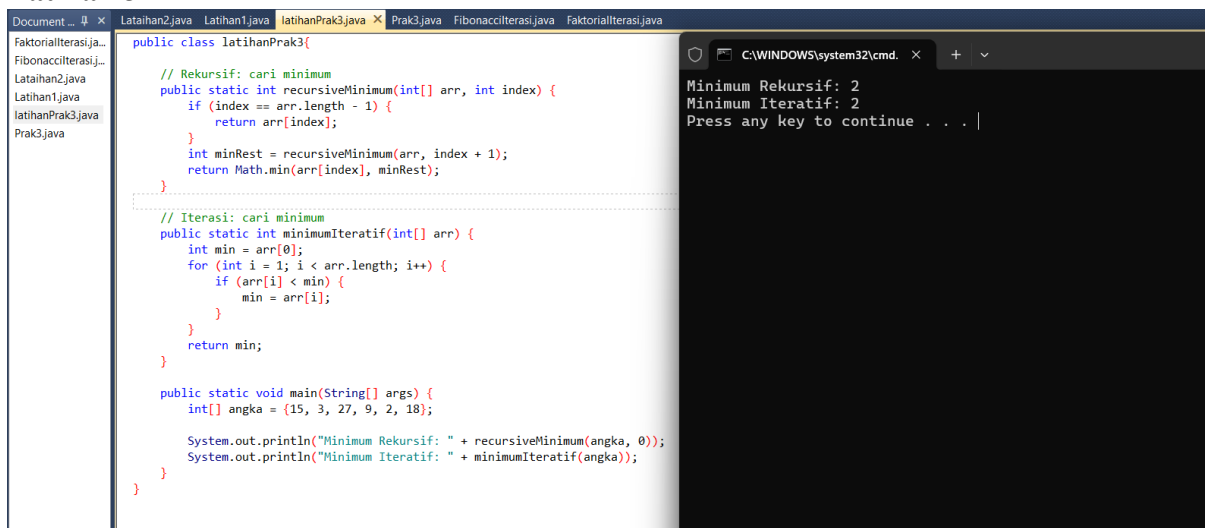
The screenshot shows an IDE with several tabs: FaktorialIterasi.java, FibonacciIterasi.java, Latihan2.java, Latihan1.java, latihanPrak3.java, Prak3.java, FibonacciIterasi.java, and FaktorialIterasi.java. The active tab is Latihan2.java, which contains the following code:

```
public class Latihan2 {  
  
    // Rekursif: reverse char array print  
    public static void stringReverseRekursif(char[] arr, int index) {  
        if (index >= 0) {  
            System.out.print(arr[index]);  
            stringReverseRekursif(arr, index - 1);  
        }  
    }  
  
    // Iterasi: reverse string print  
    public static void stringReverseIteratif(String str) {  
        char[] arr = str.toCharArray();  
        for (int i = arr.length - 1; i >= 0; i--) {  
            System.out.print(arr[i]);  
        }  
    }  
  
    public static void main(String[] args) {  
        String kata = "Praktikum";  
  
        System.out.print("Rekursif: ");  
        stringReverseRekursif(kata.toCharArray(), kata.length() - 1);  
        System.out.println();  
  
        System.out.print("Iteratif: ");  
        stringReverseIteratif(kata);  
        System.out.println();  
    }  
}
```

The terminal window on the right shows the output of the program:

```
Rekursif: mukitkarP  
Iteratif: mukitkarP  
Press any key to continue . . . |
```

Latihan 3



The screenshot shows an IDE with several tabs: FaktorialIterasi.java, FibonacciIterasi.java, Latihan2.java, Latihan1.java, latihanPrak3.java, Prak3.java, FibonacciIterasi.java, and FaktorialIterasi.java. The active tab is latihanPrak3.java, which contains the following code:

```
public class latihanPrak3 {  
  
    // Rekursif: cari minimum  
    public static int recursiveMinimum(int[] arr, int index) {  
        if (index == arr.length - 1) {  
            return arr[index];  
        }  
        int minRest = recursiveMinimum(arr, index + 1);  
        return Math.min(arr[index], minRest);  
    }  
  
    // Iterasi: cari minimum  
    public static int minimumIteratif(int[] arr) {  
        int min = arr[0];  
        for (int i = 1; i < arr.length; i++) {  
            if (arr[i] < min) {  
                min = arr[i];  
            }  
        }  
        return min;  
    }  
  
    public static void main(String[] args) {  
        int[] angka = {15, 3, 27, 9, 2, 18};  
  
        System.out.println("Minimum Rekursif: " + recursiveMinimum(angka, 0));  
        System.out.println("Minimum Iteratif: " + minimumIteratif(angka));  
    }  
}
```

The terminal window on the right shows the output of the program:

```
Minimum Rekursif: 2  
Minimum Iteratif: 2  
Press any key to continue . . . |
```

TUGAS

Tugas 1

-